

DRP Presentation: Review of Asymptotics

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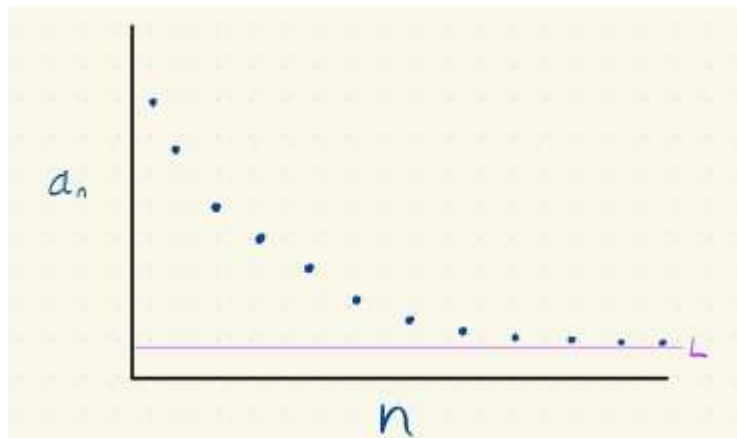
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1. Starting point: convergence of real numbers

A **sequence** is an ordered list of real numbers (denoted $\{a_n\}_{n=1}^{\infty}$ or $a_1, a_2, a_3, a_4, \dots$)

Intuitively a sequence converges to a limit L , if $\{a_n\}_{n=1}^{\infty}$ gets “really really close” to L as n gets large.



Formal definition:

a real sequence a_n converges to a limit a if for given $\epsilon > 0$, there exists some $N \in \mathbb{N}$ such that if $n > N$, then $|a_n - a| < \epsilon$

Convergence in distribution

What does it mean for a sequence of **random variables** X_n to get “closer and closer” to a limiting random variable X ?

This is formalized through “**convergence in distribution.**”

Intuitively, X_n converges in distribution to X if the cumulative distribution functions of the X_n get closer and closer to the cumulative distribution function of X .

Definition 6.1.1 (Convergence in Distribution).

A sequence of random variables $X_1, X_2, \dots$ with X_n having distribution function F_n converges in distribution to a random variable X with distribution function F if

$$F_n(x) \rightarrow F(x), \quad \text{as } n \rightarrow \infty$$

for every real number x that is a continuity point of F . We indicate this by writing

$$X_n \xrightarrow{\mathcal{D}} X, \quad \text{as } n \rightarrow \infty.$$

CLT: review of informal definition

To see an example, let us review the intuitive definition of the Central Limit Theorem (CLT):

For a sequence of random variables $X_1, \dots, X_n$ following any distribution $f(x_n)$, as sample size $n \rightarrow \infty$, $\bar{X} \approx N(\mu, \sigma)$.

This is a result that we have already used extensively in our frequentist inferential statistics approach. But we will proceed to investigate the formal definition of CLT

Formal definition of CLT

The formal definition of the CLT invokes this language of “convergence in distribution”

Theorem 6.2 (The Central Limit Theorem (CLT)). *If $X_1, X_2, \dots$ is a sequence of independent, identically distributed random variables having mean μ and variance σ^2 and*

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad (6.2)$$

is the sample mean for sample size n , then

$$\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} Y, \quad \text{as } n \rightarrow \infty, \quad (6.3)$$

where $Y \sim \mathcal{N}(0, \sigma^2)$.

This looks quite different from the informal one that we are all used to! How are the informal and formal definitions connected?

Showing the connection between formal and informal

We begin with the formal definition: $\sqrt{N}(\bar{X}_n - \mu) \rightarrow Y \sim N(0, \sigma^2)$

Recall that if $Y \sim N(\mu, \sigma^2)$, then $aY + b \sim N(a\mu + b, a^2\sigma^2)$ (by linearity of expectation and properties of variance)

Moving the $\sqrt{N}$ to the right: $(\bar{X}_n - \mu) \sim N\left(0, \frac{\sigma^2}{N}\right)$

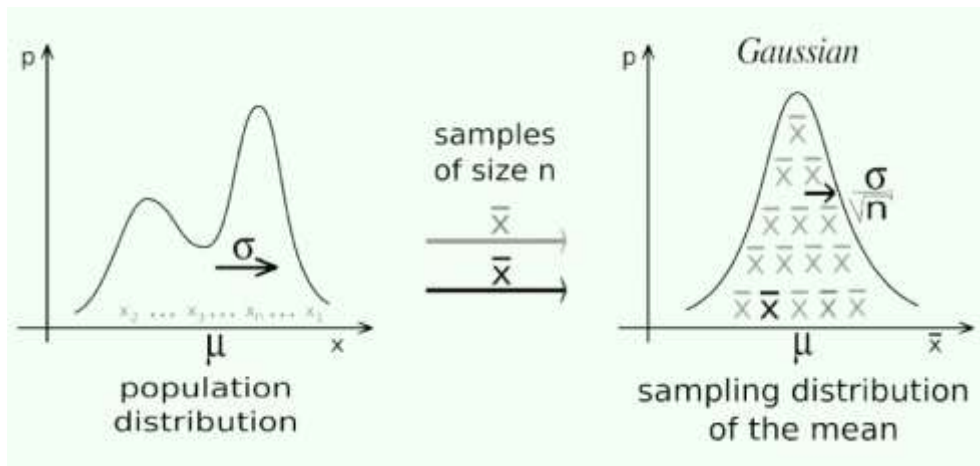
Then, moving the μ to the right-hand side by subtracting it from both sides:

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{N}\right)$$

Notice that we have arrived at the informal definition.

Connection between the informal and formal

The informal definition of CLT gives us a very nice intuition that we often do not have to think too hard about, the only condition we care about is the “sample size going to infinity” part, which in practice is encoded as a sufficiently large sample size.



Why do we need the formal version?

The formal version of CLT that we have just seen is an example of a rigorous statement, and the point of rigorous statements is to have precision in language that cannot be misunderstood. (for example, “approximately distributed as” is a vague term that could be interpreted in many different ways)

Intro to Pivot in statistics

A pivot is basically a value that does not depend on the unknown parameters (which means that it doesn't have to be a statistics.)

They are important because since they do not rely on the unknown parameters we are trying to estimate, they are quantities crucial for constructing inference such as confidence intervals.

Examples:

Recall for a normal distribution with mean μ and unknown variance σ^2 , the T-statistics

$$T = \frac{\sqrt{n}(\bar{X} - \mu)}{s}$$

Is a pivot we use to construct the confidence interval to test when population standard deviation is unknown.

A few other useful tools when dealing with asymptotics

continuous mapping theorem: arithmetic operations preserve convergence in probability
 X_n a sequence of K -dimensional random vectors.

Slutsky's theorem:

also allows us to make claims about the convergence of random variables,
(when multiplied by a sequence of real numbers, the convergence is preserved)

Portmanteau theorem:

A bunch of equivalent statements about convergence in distribution

Prokhorov's theorem:

Essentially, the random variable/metric space version of the Bolzano-Weierstrauss theorem

Delta Method:

Allows us to approximate behaviors of functions of certain random variables (Normal)

A few other useful tools when dealing with asymptotics

Delta Method:

Basic Idea:

we can approximate the asymptotic behaviour of functions over a random variable, if the random variable is itself asymptotically normal.

Given some sequence of random variables $X_1, \dots, X_n$, such that $n \rightarrow \infty$, we get $X_n \sim N(\mu, \frac{\sigma^2}{n})$.

Rearranging to get $\frac{\sqrt{n}(X_n - \mu)}{\sigma} \sim N(0, 1)$, and if given a smooth function g , the delta

Method states that: $\frac{\sqrt{n}(g(X_n) - g(\mu))}{|g'(\mu)|\sigma} \approx N(0, 1)$, which gives us $g(X_n) \approx N\left(g(\mu), \frac{g'(\mu)^2 \sigma^2}{n}\right)$